Dynamic Asset Allocation for Retirement: From Exact Dynamic Programming to Deep Reinforcement Learning

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CME 241: Reinforcement Learning for Finance
Stanford University, Winter 2026

Abstract

We study a 35-year finite-horizon retirement portfolio allocation problem formulated as a Markov Decision Process (MDP) in which an investor aged 30–65 maximizes expected CRRA utility of terminal wealth. While we provide an ideal world version of the problem, we implement simplified solutions at two levels of realism. **Phase 2** solves a simplified version via exact backward induction (dynamic programming) on a 60-bin log-spaced wealth grid with i.i.d. three-scenario returns, yielding a globally optimal policy. **Phase 3** extends to a continuous-wealth, three-state Markov regime environment solved with a Deep Q-Network (DQN) trained on sampled trajectories with switching costs and a six-action blended portfolio set. The DQN achieves a mean CRRA utility of **0.9376** vs. **0.7614** for the DP baseline (+23%), with a median terminal wealth of \$18.9M vs. \$4.9M. We further perform sensitivity analysis on on the action space, as well as ablation studies on the reward functions.

1 Introduction

Retirement investing is a canonical sequential decision problem under uncertainty. An investor allocates wealth across assets over a 35-year horizon, facing stochastic returns, evolving risk preferences, and compounding allocation mistakes. The investor's risk aversion increases with age, returns exhibit regime-dependent dynamics, and a decision made at 30 has non-linear consequences at 65. These features make the problem a natural fit for stochastic control and, specifically, Markov Decision Processes.

We develop the problem in three levels of complexity:

1. **Ideal version** — full state (income, regimes, taxes, portfolio composition), continuous multi-asset actions with transaction costs and leverage limits. This is an aspirational version that would take a long time to implement for a class project, thus we simplify it in phases 2 and 3 below.
2. **Phase 2 (DP)** — wealth discretised to 60 log-spaced bins, i.i.d. three-scenario returns, and six actions. Solved with exact backward induction.
3. **Phase 3 (DQN)** — continuous wealth, three-state Markov regimes, six-action space, and switching costs. Solved with a Deep Q-Network.

Contributions. (1) A clean empirically calibrated MDP formulation with regime-switching returns; (2) an exact DP baseline exposing a grid-discretizations artifact; (3) a DQN with regime-aware state, switching costs, and a six-action space; (4) a systematic sensitivity analysis over action-space granularity; (5) an ablation study over reward-function design including age-dependent CRRA; (6) a principled DP-vs-DQN comparison on both model richness and solver expressiveness.

2 Problem Formulation

2.1 MDP Components

The planning horizon is $T = 35$ discrete annual steps ($t = 0$ corresponds to age 30, $t = T$ corresponds to age 65). The state at time t is:

$$S_t = (A_t,\ W_t,\ M_t), \quad A_t = 30 + t,$$

where $W_t > 0$ is current wealth and $M_t \in \{\text{bull, neutral, bear}\}$ is the market regime. Wealth evolves as:

$$W_{t+1} = (W_t + c)(1 + r_t) - \delta_t, \tag{1}$$

where $c = \$10{,}000$ is the fixed annual contribution, r_t is the portfolio return, $\delta_t = 0.005\, W_t$ is a switching cost incurred when the action changes, and zero otherwise.

The **action** $a_t \in \mathcal{A}$ specifies the portfolio allocation. Regimes follow a first-order Markov chain $\mathbb{P}(M_{t+1} = m' \mid M_t = m)$ estimated from data.

2.2 Objective and Bellman Equation

The investor maximizes expected CRRA utility of terminal wealth:

$$\max_{\pi} \mathbb{E}^{\pi}[U(W_T; \gamma)] \tag{2}$$

$$U(w; \gamma) = \begin{cases} \dfrac{(w/W_{\text{ref}})^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ \ln(w/W_{\text{ref}}), & \gamma = 1. \end{cases} \tag{3}$$

with $W_{\text{ref}} = \$1{,}000{,}000$ and $\gamma = 1$ yielding $U = \ln(w/W_{\text{ref}})$. Our formulation divides by W_{ref} to keep the utility function stable, especially for studying different reward functions, which we present in the ablation studies section.

The optimal value function satisfies:

$$V_t(s) = \max_{a \in \mathcal{A}} \mathbb{E}[r_t + \beta\, V_{t+1}(S_{t+1}) \mid S_t = s,\, A_t = a]\,, \tag{4}$$

with discount $\beta = 0.99$ and terminal condition $V_T(s) = U(W_T; \gamma)$. A dense intermediate reward $r_t = \alpha\, \Delta \log W_t$ with $\alpha = 0.01$ replaces the sparse terminal utility during training, providing stable gradient signal across the 35-step horizon.

2.3 Simplification Roadmap

Table 1 summarises the deliberate simplifications from the ideal problem to each implemented version. Phase 2 makes the state space finite and the return distribution discrete, enabling exact computation. Phase 3 relaxes the two strongest approximations (wealth discretisation and regime blindness) at the cost of requiring approximate RL.

3 Data and Calibration

We use 35 annual total-return observations (1990–2024) for three Vanguard index funds from Yahoo Finance: VFINX (US equity - high risk), VBMFX (total bond - medium risk), and VFSTX (short term investment-grade - low risk). Table 2 summarises the historical return statistics. Regimes are labelled by VFINX annual return: *bull* ($> +15\%$), *neutral* (-5% to $+15\%$), *bear* ($< -5\%$), yielding 17, 13, and 5 years respectively.

Table 2: Annual total-return statistics, 1990–2024 (35 obs.).

Asset	Mean	Std	Min	Max
VFINX (Equity)	12.12%	17.36%	−37.0%	+37.8%
VBMFX (Bond)	5.17%	5.56%	−13.2%	+18.3%
VFSTX (Short IG)	4.65%	4.26%	−5.8%	+14.1%

For Phase 2, each fund's return is approximated by a three-scenario distribution (bear/neutral/bull) fitted to empirical percentile buckets. For Phase 3, each fund's return within each regime is modelled as $r_{a,m} \sim \mathcal{N}(\mu_{a,m}, \sigma^2_{a,m})$, clipped to $[-60\%, +60\%]$. The regime transition matrix (Table 3) is estimated with Laplace smoothing (+0.5 per cell). Bear regimes are empirically highly transient: the self-transition probability is only 0.38 (vs. the hand-tuned prior of 0.70), and bear $\rightarrow$ bull is the most likely next-year outcome (0.54). Full scenario distributions and regime-conditioned return parameters appear in Appendix A.

Table 3: Empirical regime transition matrix $\hat{P}$ (Laplace smoothing). Rows = from, cols = to.

	Bull	Neutral	Bear
Bull	0.49	0.37	0.14
Neutral	0.45	0.45	0.10
Bear	0.54	0.08	0.38

4 Phase 2: Exact Dynamic Programming

Setup. Wealth is discretised into 60 log-spaced bins spanning \$1k–\$5M (approximately 16.5% per bin). Six actions: three pure picks (VFSTX, VBMFX, VFINX) and three equal-weight 50/50 blends (VFSTX+VBMFX, VFSTX+VFINX, VBMFX+VFINX). The Bellman update (Eq. 4) requires $3 \times 3 = 9$ expected-value evaluations per state; with a full 35 (years) $\times$ 60 (wealth bins) = 2,100-state backward pass.

Optimal policy. VFINX dominates across all ages. The 50/50 VFSTX+VFINX blend appears in 10–22% of bins. Pure VBMFX is never selected. Figure 1 shows the full policy and value heatmaps.

Evaluation. Table 4 shows a condensed comparison across key policies (10,000 Monte Carlo paths); Figure 2 shows mean CRRA utility.

Table 1: Problem version comparison across the three-level hierarchy.

Dimension	Ideal Version	Phase 2 — DP	Phase 3 — DQN
Wealth state	Continuous	60 log-spaced bins	Continuous (norm. log-wealth)
Return model	Full distribution	3-scenario i.i.d.	Regime-conditioned Gaussian
Regimes	Markov chain	None (i.i.d.)	3-state Markov chain
Action set	Continuous weights	6 picks & blends	6 picks & blends
Switching cost	Yes	No	0.5% of W_t
Solver	—	Backward induction	DQN (mini-batch SGD)

Table 4: Phase 2 policy comparison (10,000 paths, $W_0 =$ \$100k).

Policy	Median (\$M)	P10 (\$M)	CRRA Util.
Always VFINX	**7.78**	**2.90**	**0.8284**
DP Optimal	4.93	2.72	0.7614
Glide Path	3.39	1.71	0.6621
Always VBMFX	1.53	1.16	0.3318
Always VFSTX	1.35	1.10	0.2503

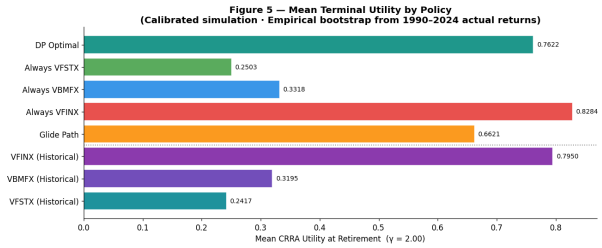

Figure 2: Mean CRRA utility (Phase 2, $\gamma = 1$) for all modelled and empirical policies. The DP Optimal policy ranks second; Always VFINX performed best.

$$y_t = r_t + \beta \max_{a'} Q_{\theta^-}(s_{t+1}, a') \cdot (1 - \mathbf{1}_{\text{done}}),$$

where θ^- is the periodically frozen target network.

Training. 5,000 episodes; replay buffer 50,000; target network updated every 500 episodes; ε-greedy with $\varepsilon : 1.0 \to 0.05$ over ≈590 episodes. Figure 3 shows the training curve; mean episode reward rises from 0.0368 (ep 500) to 0.0495 (ep 5,000), indicating stable convergence. Full hyperparameters in Appendix C.

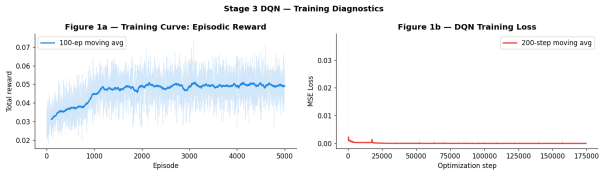

Figure 3: DQN training curve: raw episode reward (light) and 100-episode moving average (dark). Reward stabilises after ε reaches its minimum at ≈episode 590.

5 Phase 3: Deep Q-Network

State encoding. The 5-dimensional normalised state vector is:

$$\mathbf{s}_t = \left[\frac{t}{35},\ \frac{\log_{10} W_t}{\log_{10} W_{\max}},\ \mathbf{1}_{\text{bull}},\ \mathbf{1}_{\text{neutral}},\ \mathbf{1}_{\text{bear}}\right].$$

Log-wealth normalisation is common for exponentially growing quantities; one-hot regime encoding avoids imposing a numeric ordering on regimes.

Action space (Scheme B). Six actions: three pure picks (VFSTX, VBMFX, VFINX) and three equal-weight 50/50 blends (VFSTX+VBMFX, VFSTX+VFINX, VBMFX+VFINX). Scheme B was selected via sensitivity analysis (Section 7).

Network. A fully connected MLP: $5 \to 64 \to 64 \to 6$ with ReLU activations (4,934 parameters). The loss is:

$$\mathcal{L}(\theta) = \mathbb{E}\left[\left(Q_\theta(s_t, a_t) - y_t\right)^2\right],$$

Results. Table 5 shows the main evaluation (10,000 episodes). Figure 4 shows mean CRRA utility across all policies. The DQN achieves the highest median wealth and CRRA utility of all policies. Notably, its P10 of \$9.17M versus Always VFINX's P10 of \$1.43M demonstrates strong downside protection — the regime-conditioned switching between equity and bonds cushions bear-market losses. The full nine-policy table (including empirical bootstrap baselines) appears in Appendix C.

Table 5: Phase 3 policy comparison (10,000 paths, $W_0 =$ \$100k).

Policy	Med. (\$M)	P10 (\$M)	CRRA Util.
DQN	**18.87**	**9.17**	**0.9376**
Always VFINX	5.64	1.43	0.6757
Glide Path	2.24	1.06	0.4750
Always VBMFX	1.50	1.06	0.3088
Always VFSTX	1.32	1.01	0.2263

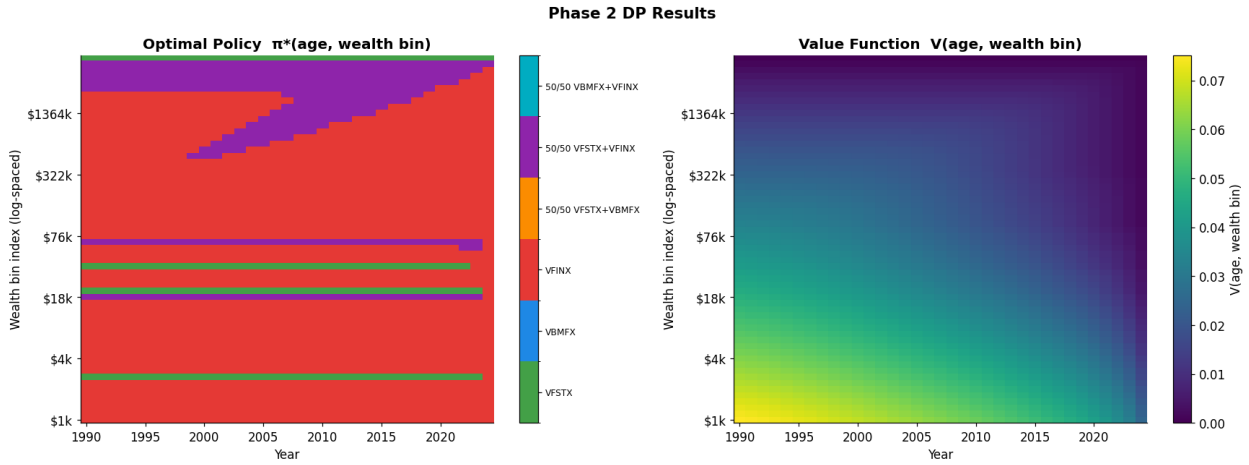

Figure 1: Phase 2 DP results. **Left:** optimal policy heatmap $\pi^*(t, w_i)$ — VFINX (red) dominates most bins; 50/50 VFSTX+VFINX (purple) appears at medium-to-high wealth in mid-career. **Right:** value function $V(t, w_i)$; values are larger early and at low wealth, reflecting longer compounding horizon.

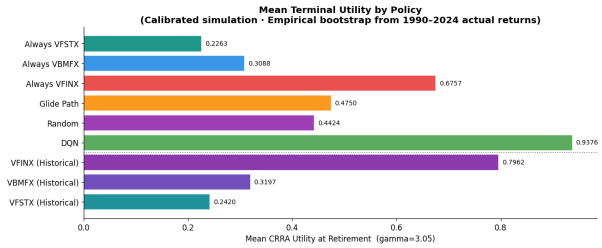

Figure 4: Mean CRRA utility (Phase 3, $\gamma = 1$) for all evaluated policies. DQN achieves the highest utility.

6 DP vs. DQN: Why Deep RL Is Superior

The DQN outperforms the DP on every metric (+23% CRRA utility, +$14M median wealth, substantially higher P10). The gap has two sources: a richer environment and a more expressive solver.

1. State representation. DP bins wealth into 60 cells (≈16.5% spacing per bin), introducing systematic approximation error. The DQN operates on continuous normalised log-wealth; its neural Q-function can represent arbitrary value gradients within a wealth range.

2. Return model fidelity. DP uses i.i.d. three-scenario returns. DQN employs regime-conditioned Gaussians with an empirically estimated Markov transition matrix, capturing the documented persistence of equity bull and bear phases.

3. Regime awareness. The DQN observes M_t and can condition its allocation on the current regime (e.g. reduce equity exposure mid-bear). The DP has no regime variable and is blind to this signal.

4. Transaction friction. DP ignores switching costs entirely. The DQN's 0.5%-of-wealth penalty discourages churning, producing smoother, more realistic policies.

5. Scalability. Adding a fourth asset or a second regime dimension multiplies the DP state space; the neural Q-function accommodates it with a modest change in input dimension.

7 Sensitivity Analysis: Action Space

We compare three action-space granularities to select the optimal scheme for the DQN (Table 6). All variants are trained for 5,000 episodes (seed 42) and evaluated over 2,000 episodes.

Table 6: Action-space schemes evaluated.

Scheme	# Actions	Description
A	3	100% VFSTX; 100% VBMFX; 100% VFINX
B	6	Scheme A plus three 50/50 blends
C	15	All (w_0, w_1, w_2), $w_i \in \{0, 0.25, \ldots, 1\}$, $\sum w_i = 1$

Table 7: Sensitivity results (2,000 eval. episodes each).

Scheme	Median TW ($M)	CRRA Util.	Convergence
A (3 pure picks)	15.66	0.4843	Smooth, fastest
B (6: pure+blends)	**18.57**	**0.4854**	Smooth
C (15: 25% grid)	4.02	0.4482	Noisy, unstable

Scheme B achieves the highest median terminal wealth ($18.57M) and mean CRRA utility (0.4854). Scheme A converges fastest but misses diversification gains from blended portfolios. Scheme C fails to converge cleanly within 5,000 episodes: the 15-action space causes the curse of exploration, spreading ε-greedy experience too thinly across actions. Figure 5 shows summary bars and training curves; additional distribution and allocation plots appear in Appendix D.

8 DQN Ablation Study: Reward Function Design

We isolate the effect of the DQN training reward signal by holding all other hyperparameters fixed (Scheme B, 10,000 training episodes, seed 42) and varying only the per-step rgeward formula. Table 8 defines the five variants.

Table 8: Reward variants. All use Scheme B.

Variant	Reward formula r_t
Baseline	$0.01 \cdot (\log W_{t+1} - \log W_t)$
Abl. B	$U(W_{t+1};\gamma_t) - U(W_t;\gamma_t),\ \ \gamma_t = 1.0 + 0.03(t-30)$
Abl. C	$U(W_{t+1};2) - U(W_t;2)$
Abl. D	$W_{t+1} - W_t$ (risk-neutral, $\gamma = 0$)
Abl. E	$U(W_{t+1};3) - U(W_t;3)$ (high risk aversion)

Table 9: Ablation evaluation (10,000 paths, W_{ref} = $1M). Wealth in $M.

Variant	Med. ($M)	Std ($M)	P10 ($M)
Baseline (log)	18.87	11.78	9.17
Abl. B (age-CRRA)	18.49	7.69	8.87
Abl. C ($\gamma = 2$)	14.91	4.05	8.51
Abl. D ($\gamma = 0$)	17.95	14.71	7.43
Abl. E ($\gamma = 3$)	3.35	1.53	2.59

Key findings. Baseline (with γ= 1) achieves the highest median wealth of $18.87M followed closely by ablation b (age-based CRRA) with a median wealth of $18.49M. However, we note that ablation B standard deviation is about $4M lower than the baseline, providing more stable returns in general. Ablation C ($\gamma = 2$ constant) gives terminal median wealth and standard deviation at $14.91M and $4.05M, respectively - indicating a more conservative investment approach. Ablation E (with γ= 3) gives the lowest median terminal wealth and the lowest standard deviation. This shows that over-penalizing downside risk forces the agent into low-return assets, sacrificing 35 years of equity compounding. Ablation D (risk-neutral), gives a median terminal wealth of $17.95M but a much bigger standard deviation of $14.71M. Overall, a balance strategy could be achieved by using the reward function in ablation B (age-based CRRA), which gives nearly optimal median terminal wealth with a relatively lower standard deviation. CRRA utility values are not compared because γ values are different in each ablation, thus giving different interpretation for each.

9 Conclusion

We have demonstrated a principled two-stage approach to the retirement asset allocation problem. Exact dynamic programming on a simplified discrete model establishes an interpretable model. A Deep Q-Network on the richer continuous-wealth, regime-aware environment substantially improves CRRA utility and terminal wealth, while also providing stronger downside protection. The sensitivity analysis identifies Scheme B (6-action) as the optimal action-space granularity. The ablation study shows that an age-varying CRRA step reward (Ablation B) provides the optimal risk-adjusted outcomes, though the simpler log-wealth shaping baseline remains highly competitive.

Future work. Continuous-action solvers (SAC) would replace Scheme B with portfolio weight vectors on the simplex. We can also increase the complexity of our model by incorporating richer state variables like stochastic labor income, CPI inflation, capital gains taxes, and post-retirement consumption (all of which would affect an individual's investment decisions). We can also use a double DQN where we ask the online network to pick the action and the offline network to evaluate it, thus reducing overestimation.

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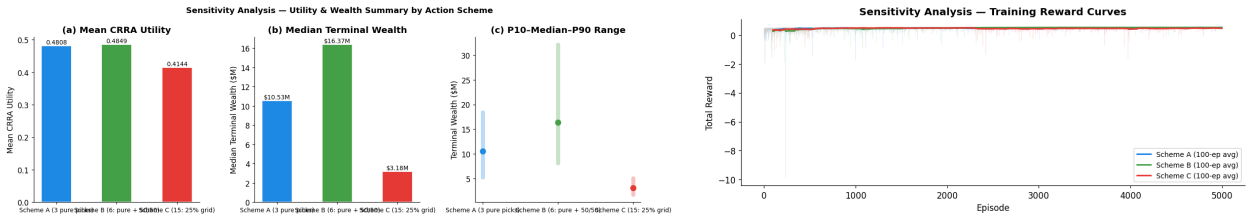

(a) Summary: median terminal wealth and mean CRRA utility by scheme.

(b) Training reward curves (100-ep moving avg.) by scheme.

Figure 5: Action-space sensitivity analysis. Scheme B (6 actions) dominates on both metrics; Scheme C training curves are noisy and never stabilise.

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A Data and Calibration Details

A.1 Phase 2: Calibrated Three-Scenario Return Distributions

Returns are split into percentile buckets: bottom 29% → bear, middle 31% → neutral, top 40% → bull.

Table 10: Phase 2 three-scenario distributions (prior hand-tuned vs. empirically calibrated).

Asset	Scenario	Prior Return	Prior Prob.	Fitted Return	Fitted Prob.
VFINX	Bear	−15.0%	0.28	−10.1%	0.29
	Neutral	+10.0%	0.32	+11.7%	0.31
	Bull	+27.0%	0.40	+28.4%	0.40
VBMFX	Bear	+10.0%	0.28	−1.4%	0.29
	Neutral	+4.0%	0.32	+4.9%	0.31
	Bull	+2.0%	0.40	+10.0%	0.40
VFSTX	Bear	+1.0%	0.28	−0.1%	0.29
	Neutral	+3.0%	0.32	+4.0%	0.31
	Bull	+4.5%	0.40	+8.5%	0.40

Blended action scenarios (50/50 weighted averages):

Table 11: Phase 2 blended action scenarios.

Blend	Bear (0.29)	Neutral (0.31)	Bull (0.40)	Mean
50/50 VFSTX+VBMFX	−0.7%	+4.5%	+9.3%	4.91%
50/50 VFSTX+VFINX	−5.1%	+7.9%	+18.4%	8.39%
50/50 VBMFX+VFINX	−5.7%	+8.3%	+19.2%	8.65%

A.2 Phase 3: Regime-Conditioned Return Parameters

Table 12: Phase 3 regime-conditioned return parameters (prior vs. empirical).

Asset	Regime	Prior μ	Prior σ	Empirical μ	Empirical σ
VFSTX	Bull	+4.5%	0.5%	+5.9%	4.0%
	Neutral	+3.0%	0.5%	+3.9%	3.2%
	Bear	+1.0%	0.5%	+2.2%	7.0%
VBMFX	Bull	+2.0%	4.5%	+5.7%	5.5%
	Neutral	+4.0%	4.5%	+4.9%	4.0%
	Bear	+10.0%	5.5%	+4.1%	9.9%
VFINX	Bull	+27.0%	15.0%	+26.3%	6.3%
	Neutral	+10.0%	15.0%	+5.9%	6.2%
	Bear	−15.0%	19.0%	−19.7%	11.0%

B Phase 2: Full Evaluation Results

B.1 DP Policy Strategy Frequency by Age

Table 13: Fraction of the 60 wealth bins assigned to each strategy by age (Phase 2 DP optimal policy). S = VFSTX, B = VBMFX, E = VFINX.

Age	VFSTX	VBMFX	VFINX	50/50 S+B	50/50 S+E	50/50 B+E	Dominant
30	6.7%	0.0%	81.7%	0.0%	11.7%	0.0%	VFINX
35	6.7%	0.0%	81.7%	0.0%	11.7%	0.0%	VFINX
40	6.7%	0.0%	78.3%	0.0%	15.0%	0.0%	VFINX
45	6.7%	0.0%	73.3%	0.0%	20.0%	0.0%	VFINX
50	6.7%	0.0%	71.7%	0.0%	21.7%	0.0%	VFINX
55	6.7%	0.0%	76.7%	0.0%	16.7%	0.0%	VFINX
60	6.7%	0.0%	83.3%	0.0%	10.0%	0.0%	VFINX

VFINX dominates throughout; the 50/50 VFSTX+VFINX blend peaks at age 50, reflecting a modest mid-career risk reduction. Pure VBMFX and 50/50 VBMFX+VFINX are never selected; pure VFSTX occupies only the lowest 4 bins (wealth $\lesssim$ \$24k).

B.2 Phase 2 Figures

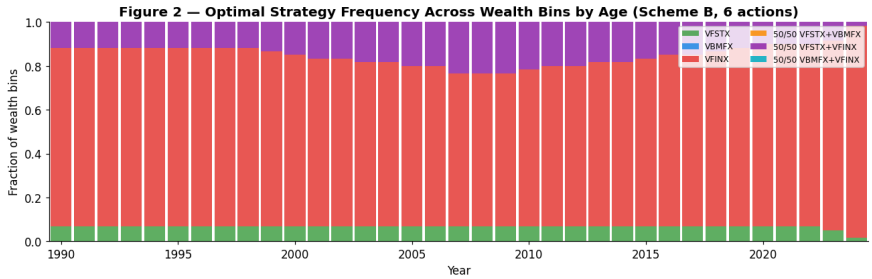

Figure 6: Phase 2 DP: fraction of wealth bins assigned to each strategy by year. VFINX (red) dominates throughout; the 50/50 VFSTX+VFINX blend (purple) peaks near age 50.

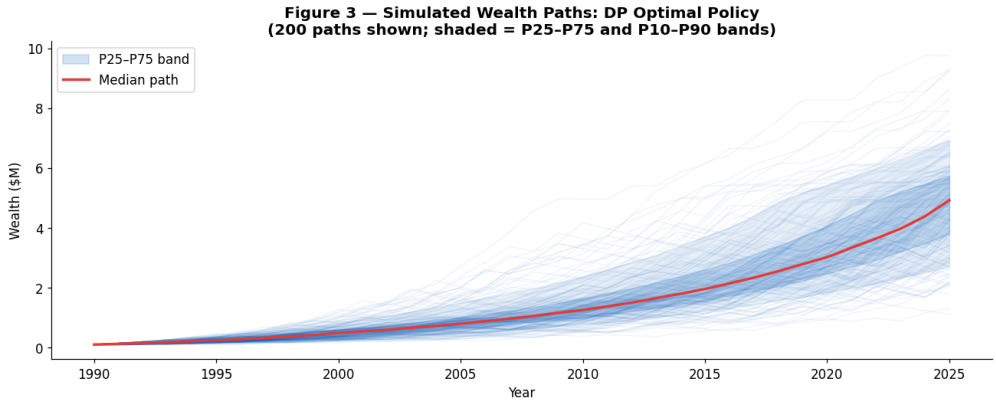

Figure 7: Phase 2 DP: 200 simulated wealth trajectories under the optimal policy, with P10–P90 and P25–P75 shaded bands. Median wealth at retirement exceeds \$4.9M.

C Phase 3: Full Evaluation Results and Hyperparameters

C.1 DQN Hyperparameters

Table 14: DQN training hyperparameters.

Hyperparameter	**Value**
Training episodes	10,000
Batch size	128
Replay buffer capacity	50,000
Learning rate (Adam)	10^{-3}
Discount factor β	0.99
ε start	1.0
ε end	0.05
ε decay	$\times$0.995 per episode
Target network update freq.	every 500 episodes (hard copy)
Hidden layer width	64
Gradient clip norm	10.0
Random seed	42

ε reaches 0.05 at episode $\approx$590 (since $0.995^{590} \approx 0.05$) and remains flat for the remaining $\approx$4,410 training episodes.

C.2 Ablation B Age-Dependent Risk Aversion Schedule

Table 15: CRRA risk aversion γ by age: $\gamma(A) = 2.0 + 0.03(A - 30)$.

Age	γ
30	2.000
35	2.150
40	2.300
45	2.450
50	2.600
55	2.750
60	2.900
65	3.050

C.3 Phase 3 Figures

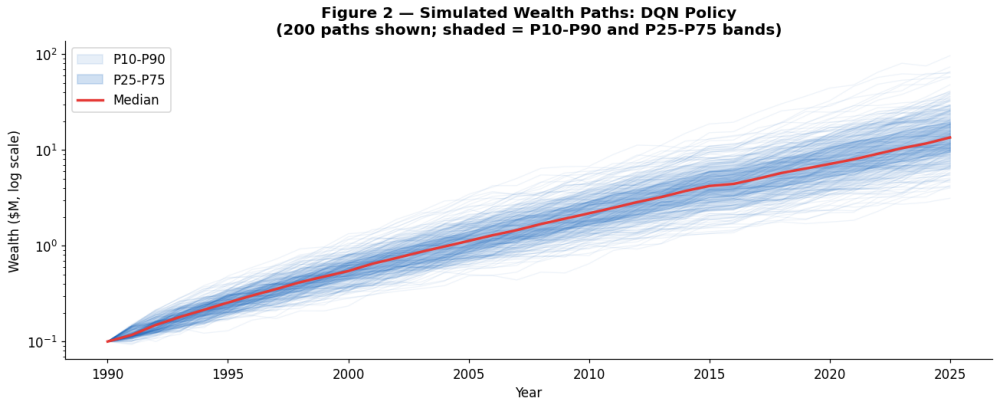

Figure 8: Phase 3 DQN: 200 simulated wealth trajectories with P10–P90 and P25–P75 shaded bands. Median terminal wealth exceeds $18.5M, substantially above any baseline.

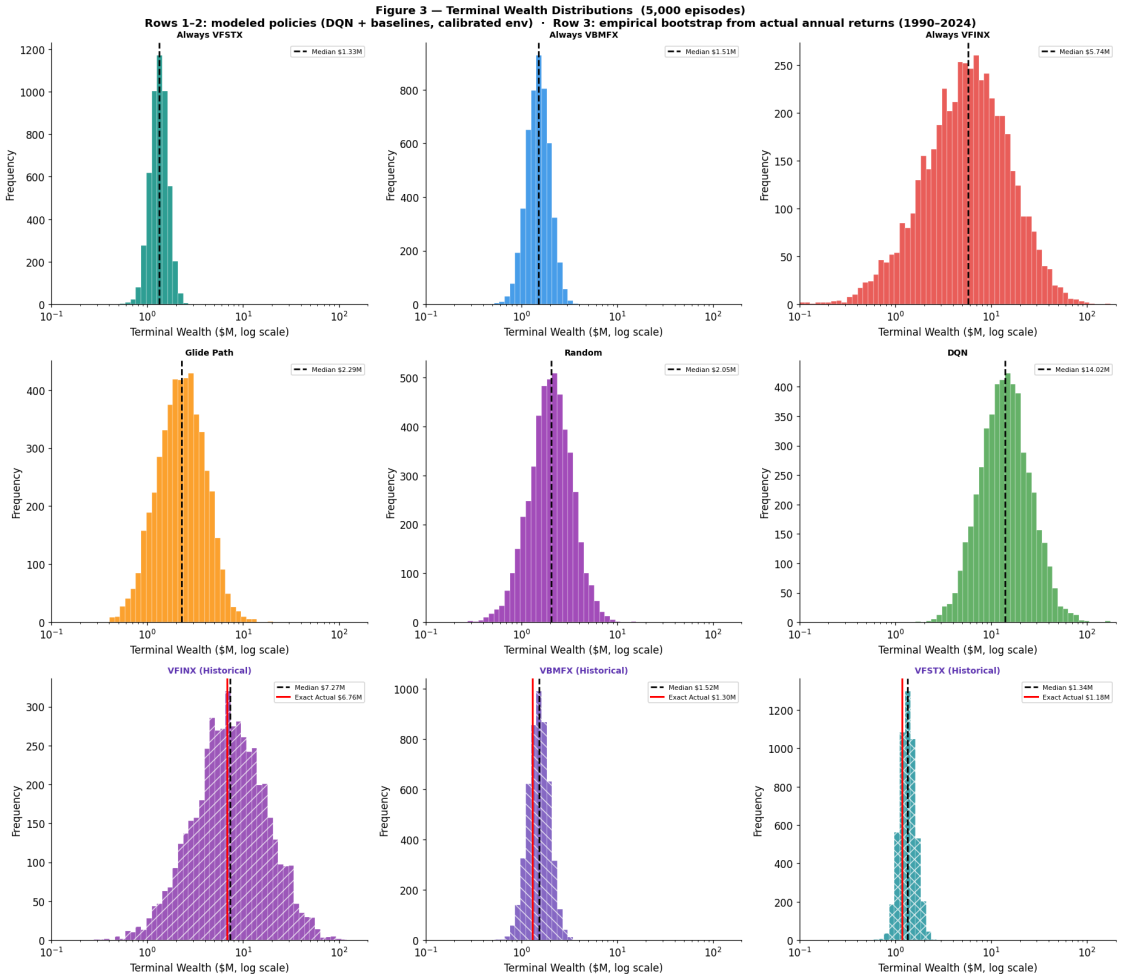

Figure 9: Phase 3: terminal wealth distributions for all evaluated policies. DQN distribution is shifted far to the right; Always VFSTX and Always VBMFX cluster near $1–2M.

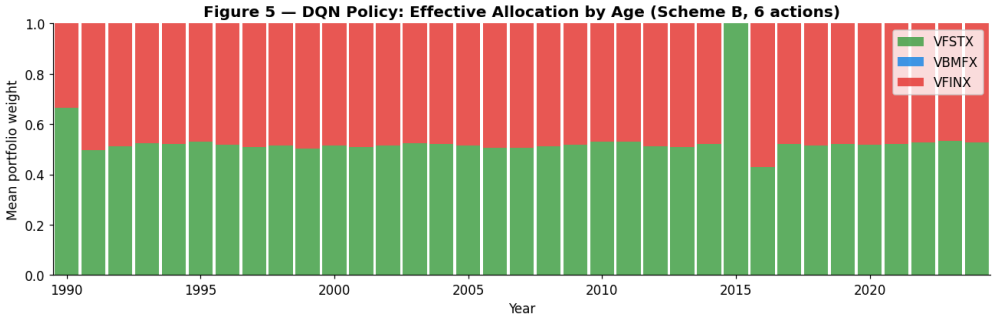

Figure 10: Phase 3 DQN: effective VFSTX/VBMFX/VFINX allocation by age (averaged over 10,000 episodes). VFINX dominates early career; bond/savings weight increases gradually toward retirement.

D Additional Sensitivity Figures

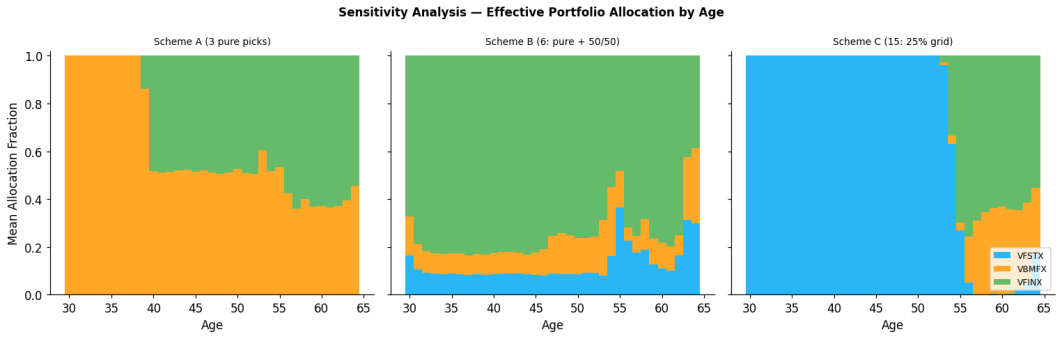

Figure 11: Sensitivity analysis: effective VFSTX/VBMFX/VFINX allocation by age for each scheme. Scheme B shows richer diversification patterns compared to Scheme A; Scheme C produces unstable allocation patterns.

E Additional Ablation and Gamma-Sensitivity Figures

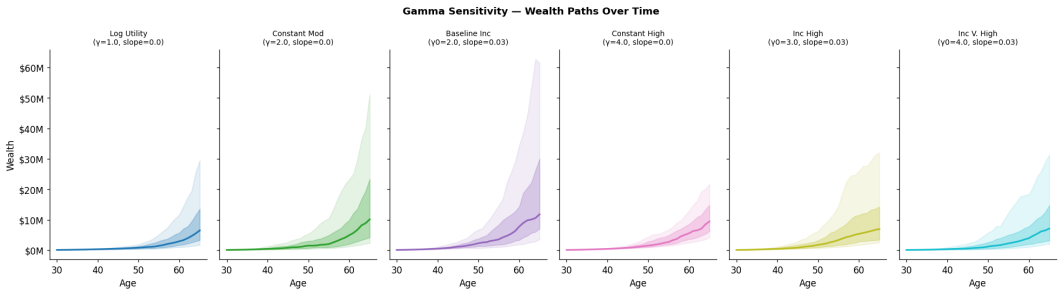

Figure 12: γ-schedule sensitivity: median wealth trajectories $\pm$ IQR by schedule design. The age-varying schedule achieves the optimal median wealth while maintaining tighter bands than the fixed-low ($\gamma = 2$) schedule.

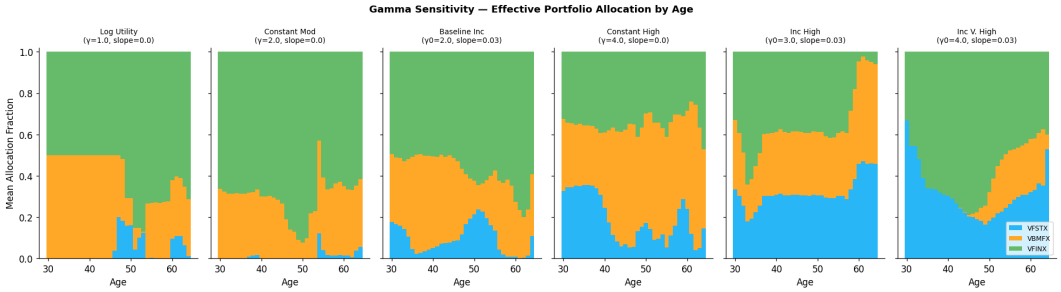

Figure 13: γ-schedule sensitivity: effective VFSTX/VBMFX/VFINX allocation by age. Fixed-high γ drives excessive early-career conservatism; the age-varying schedule transitions smoothly from equity to safer assets as retirement approaches.